Technische Hochschule Augsburg

STATISTICS

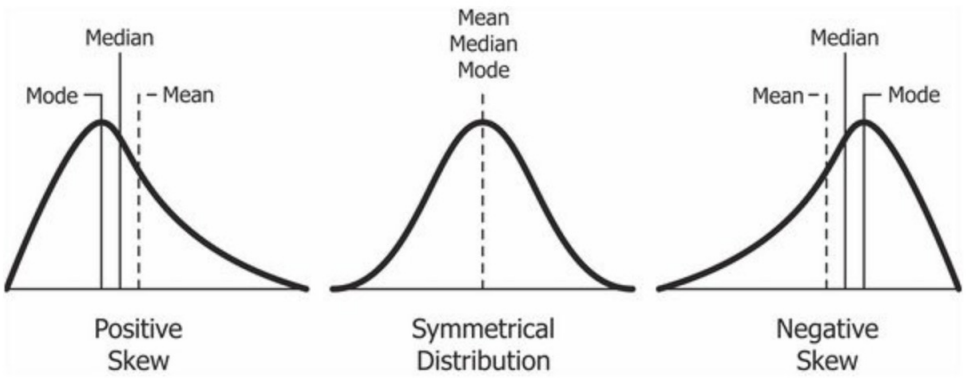
International Information Systems, 3rd semester

(5) Descriptive statistics: Visualizations

PROF. DR. -PHIL ALESSANDRA ZARCONE

LAST WEEK

Descriptive statistics



- The purpose of **descriptive statistics** is to **summarize** and **visualize** the data and understand patterns
  - show the **central tendency** with **mean, median** and **mode**
  - look at the **spread** or **dispersion** (**range**, **interquartile range**, **variance**, **standard deviation**)
  - locate and possibly remove **outliers**

Questions?

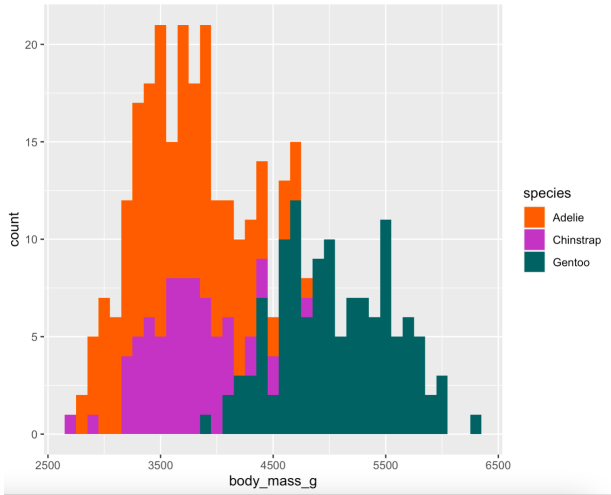
Homework solutions in **R_exercise_2.R** & **R_exercise_4.R**

PRELIMINARY COURSE PLAN

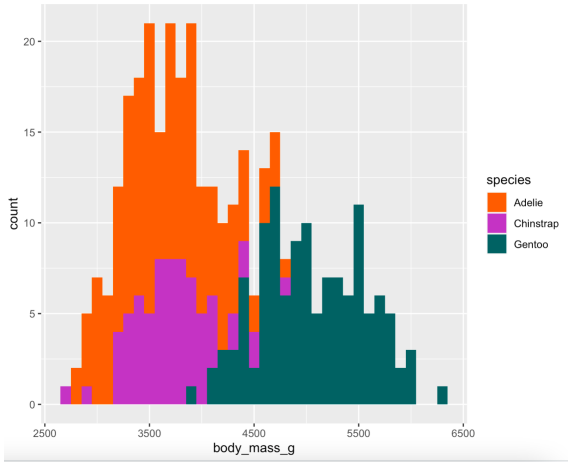
- Introduction, Descriptive statistics: measures of central tendency
- **[Oct 9th-16th] Introduction to R, Descriptive statistics: measures of dispersion, distributions, visualizations**
- [Oct 23rd] Basic probability theory
- [Oct 30th] Introduction to inferential statistics: sampling, hypothesis testing
- [Nov 6th] Test Statistics (I)
- [Nov 13th] **No class *(Tag der Lehre)***
- [Nov 20th-Nov 27th] Test statistics (II-III)
- [Dec 4th] Introduction to modeling: correlation
- [Dec 11th-18th] The Generalized Linear Model
- [Jan 9th-29th] Extra session if needed, Mock Exam, QA, extra exercises

TODAY'S TOPICS AND LEARNING GOALS

- **Visualizations**
  - choose the right way to visualize your data and create data visualizations
- **Outliers**
  - summing up all the definitions of outliers
- **Visualizations in R**
  - use R to create simple visualizations



VISUALIZATIONS



WHY DO WE NEED VISUALIZATIONS?

- The purpose of **descriptive statistics** is to summarize the data and understand patterns
  - You can use ***measures of central tendency***
    to understand around what values you can find most of your data
  - You can use ***measures of dispersion***
    to understand to what degree the values are distributed around the mean
  - You can use ***visualizations*** to learn about the shape of your data and summarize it

WHY DO WE NEED VISUALIZATIONS?

Visualizations

- are ***very useful first step*** to look at your data before you get to actually analyzing it
- ***show essential information about the data*** (to yourself, and later to an audience)

A good visualization should

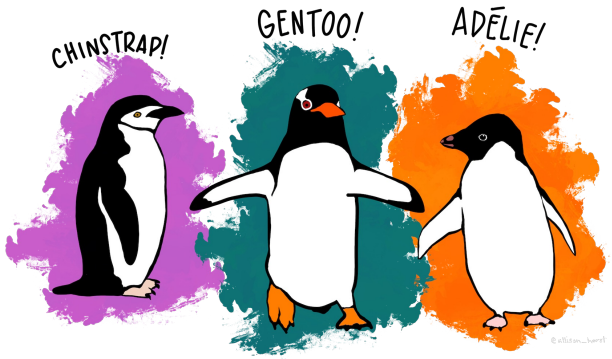
- induce the reader to think ***about the data***
- ***avoid distorting*** the data
- provide ***comparisons***

FREQUENCY TABLES

Visualizing categorical variables

- If you are dealing with ***categorical variables*** (measured on a nominal scale), you can mostly count the ***frequencies*** of the different values and collect them in a **frequency table**
  - To get a **relative frequency table**, divide each frequency by the total of datapoints (and multiply by 100)

Specie	Frequency	Relative Frequency
Adélie	152	44%
Chinstrap	68	20%
Gentoo	124	36%

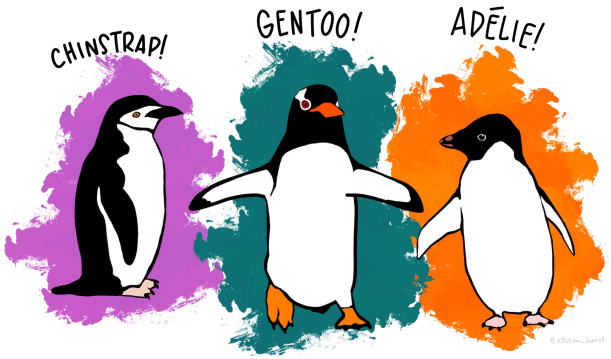


CONTINGENCY TABLES

Visualizing two categorical variables

- A **contingency table** allows you to look at how two categorical variables are combined, showing frequencies for every combination of values of the two variables

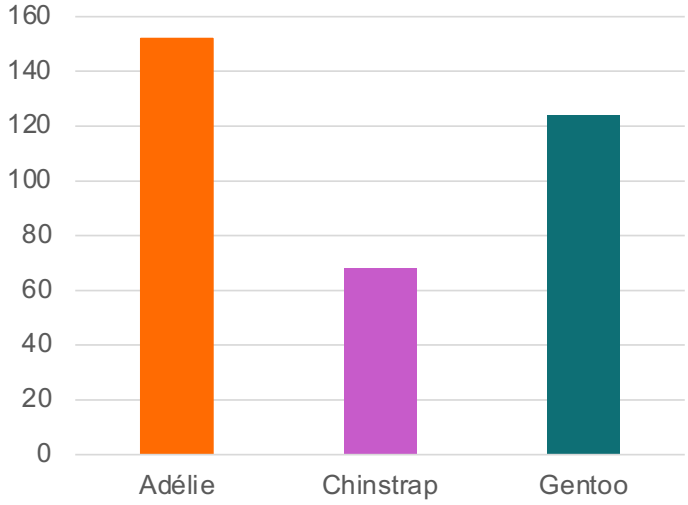
	Briscoe	Dream	Torgersen
Adélie	44	56	52
Chinstrap	0	68	0
Gentoo	124	0	0

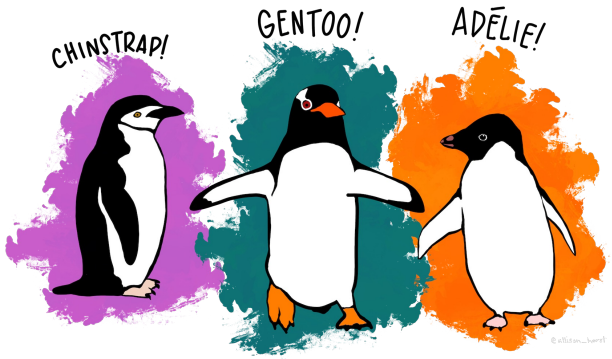


BAR CHARTS

Visualizing categorical variables

- Another way to visualize frequencies of categorical data is to use bar charts
  - each bar corresponds to a value of a variable
  - the bar height corresponds to the variable frequency

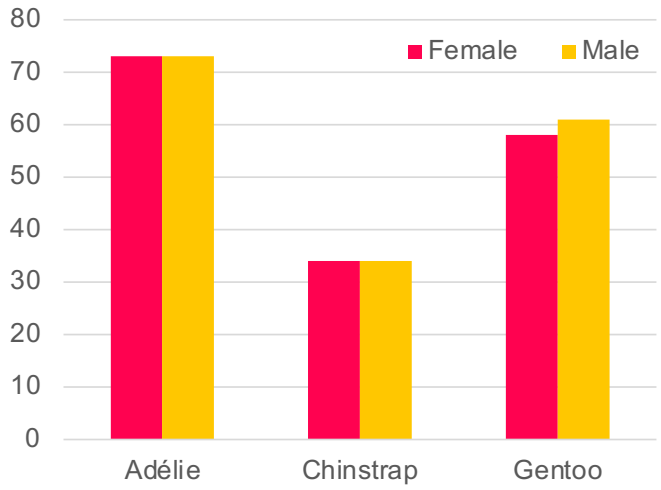


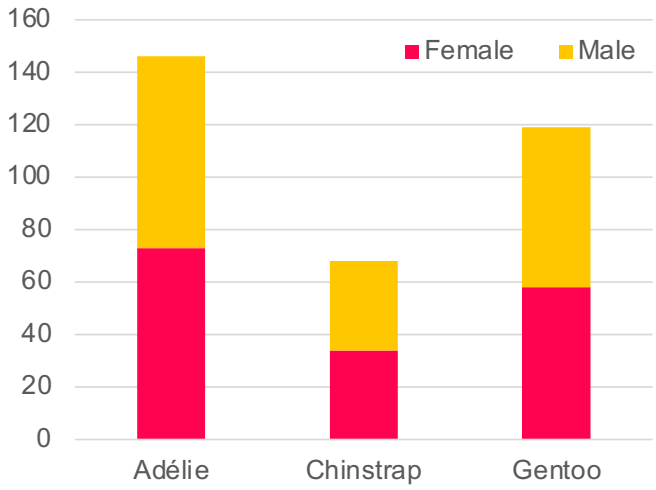


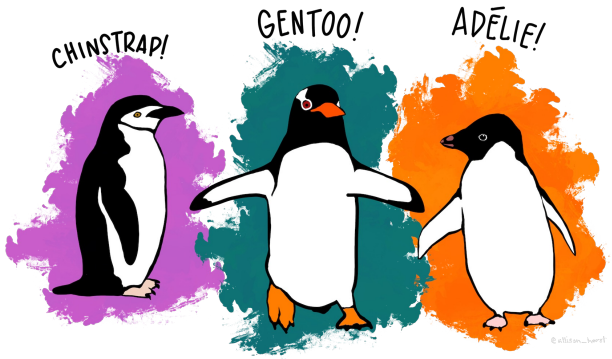
BAR CHARTS WITH TWO VARIABLES

Visualizing two categorical variables

- You can also visualize two variables at the same time on a bar chart
  - the frequencies for the different values of a category can be presented side-by-side
  - or stacked



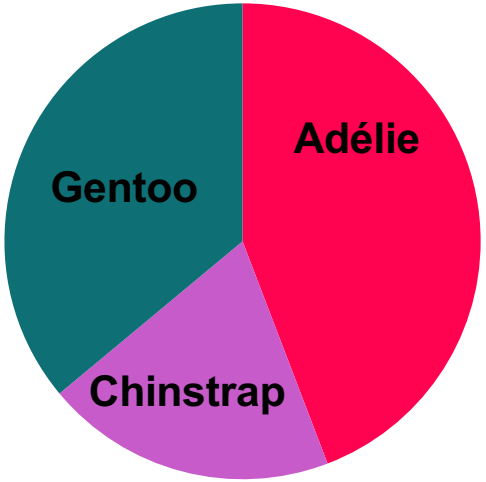


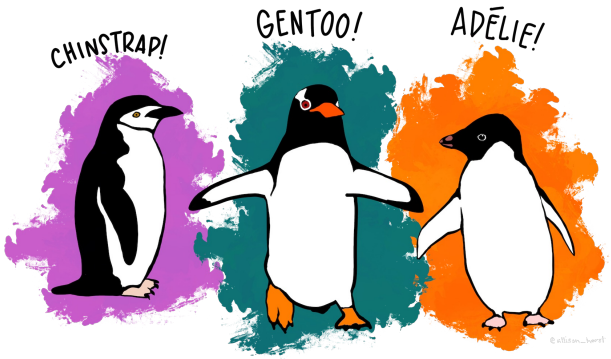


PIE CHARTS

Visualizing categorical variables

- Pie charts portion out pieces of a circle based on the relative frequency of different values
  - they are not a good way to display more than one variable at a time
  - they do not help us make an intuitive association between space (area) and size (bar plots are better for this)
  - (More on this here: The issue with pie charts)



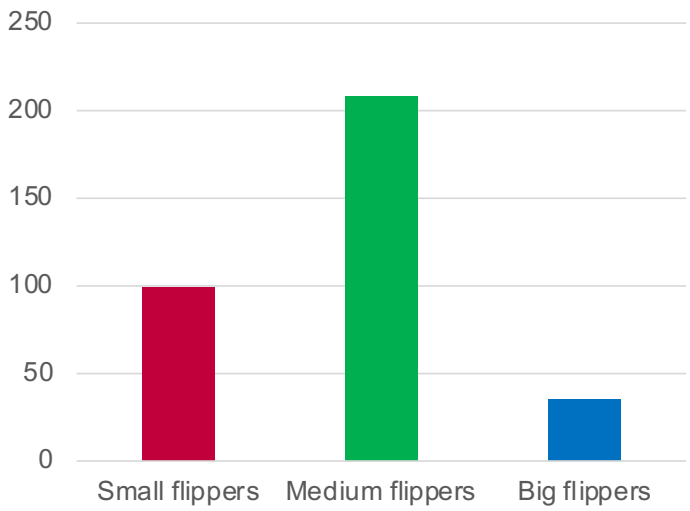


BINNING

Creating tables for continuous variables

- If you are dealing with ***continuous variables,*** you can use them to create ***categories*** too (**bins**) and treat them in the same way as categorical variable
  - The bins should ideally make some sense (have **natural cutoff points**) e.g. you can bin ages into children, teenagers, adults, elderly
  - ***Always ask yourself what the reason behind the binning is! (how did we decide what a "medium flipper size" is?)***

Flipper length	Frequency
Small flippers	99
Medium flippers	208
Big flippers	35

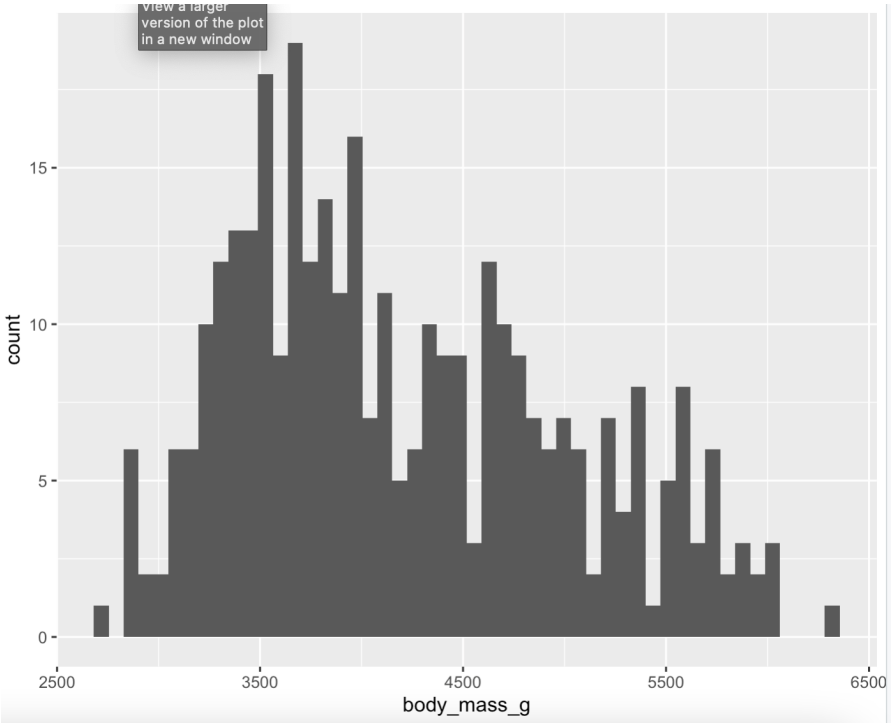


HISTOGRAMS

Visualizing continuous variables

The typical way to represent **continuous data** is with **histograms**

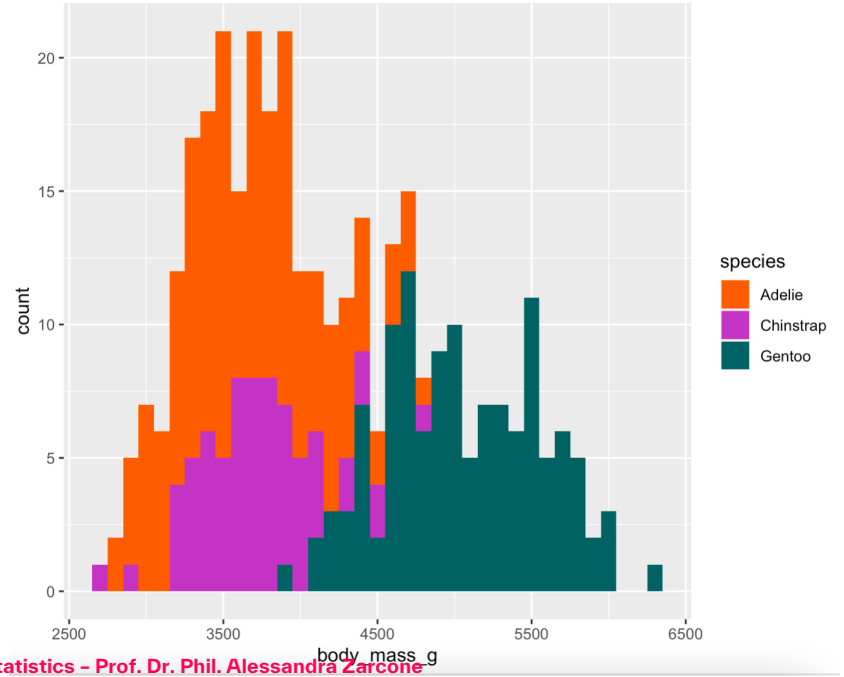
- they look like bar charts, but there is no separation between the bars (the data is ***continuous***)
- histograms show the **distribution** of continuous data

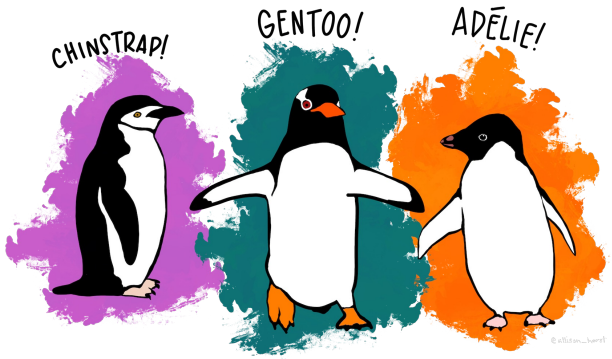


HISTOGRAMS

Visualizing continuous and categorical variables together

- A way to looking into **two variables (one categorical, one continuous)** at the same time on histogram
  - Each group (value of the categorical variable) can be highlighted in a different color





BOX-AND-WHISKERS PLOT

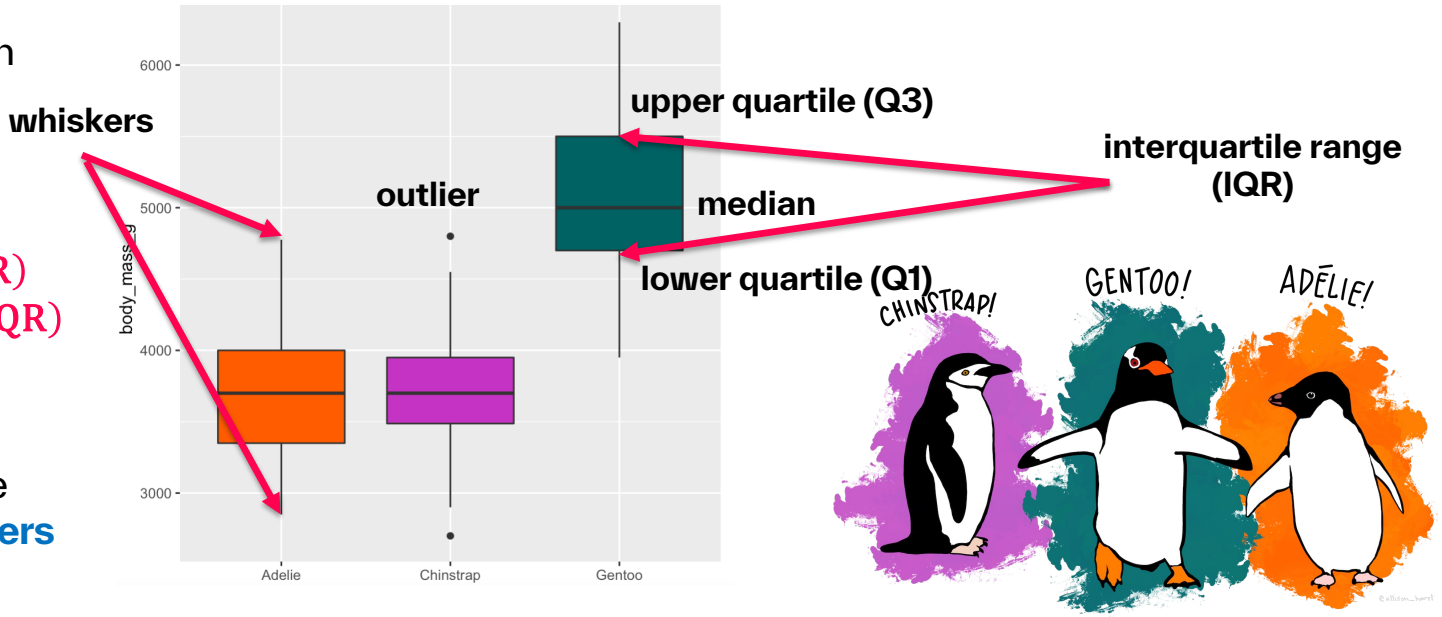
Visualizing continuous and categorical variables together

- **Boxplots** or **box-and-whiskers plot** can also show two variables at the same time (one categorical, one continuous)
  - they are a good visual representation of the **dispersion** of a sample

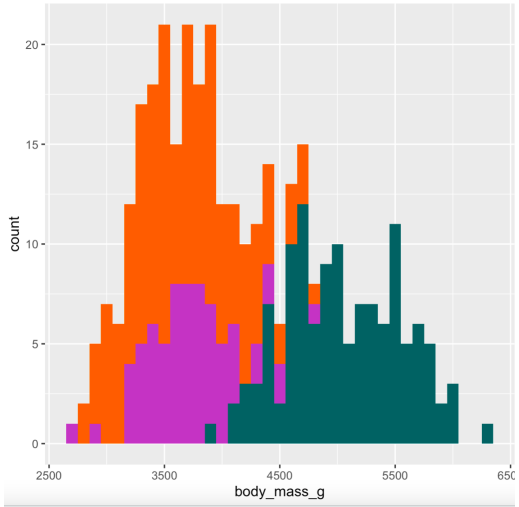
whiskers show the range between a value that's 1.5 IQR from Q1 and a value that's 1.5 IQR from Q3

$$\text{top whisker} = Q3 + (1.5 * IQR)$$
$$\text{bottom whisker} = Q1 - (1.5 * IQR)$$

datapoints that are beyond the whiskers can be considered **outliers**

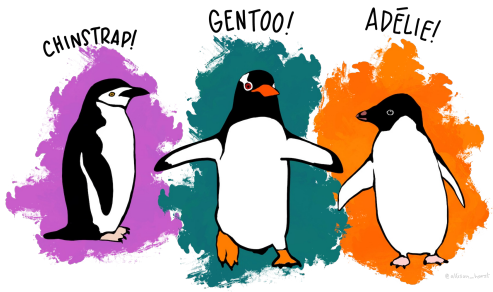


SUMMING UP (1)



- The purpose of **descriptive statistics** is to **summarize and visualize** the data and understand patterns
- *Besides* central tendency, we need to look at the **spread** or **dispersion**, that is the degree to which individual data points ***are distributed around the mean***
  - The **range**, the **interquartile range**, the **variance**, the **standard deviation** are typical measures of dispersion
  - Looking at how much scores or datapoints ***deviate*** from the mean provides us with useful information to learn about the data spread and identify **outliers**
- Tables and visualizations are a ***very useful first step*** to look at your data before you get to actually analyzing it
  - Categorical and continuous variables can be represented in different ways
  - If needed, continuous variables can be binned into categories
  - Tables and visualization can help you look at differences between groups and at relationships between variables

SUMMING UP (2)



type of variable	descriptive statistics	tables and visualizations
one, categorical (or one continuous – binned)	frequency (counts), mode	frequency and relative frequency tables, bar plots
two, categorical (or one categorical, one continuous – binned)	frequency (counts)	contingency tables for every combination of values, bar plots (binned or stacked)
one, continuous	measures of central tendency and spread	histograms
one categorical and one continuous	measures of central tendency and spread per group	histograms with colors for groups box-and-whiskers plots

OUTLIERS

HOW CAN WE IDENTIFY AND REMOVE OUTLIERS?

outliers
unusually extreme values, which stand out from the rest of the distributions

- **Cutoff values**
  removing values higher or lower than a specific cutoff value

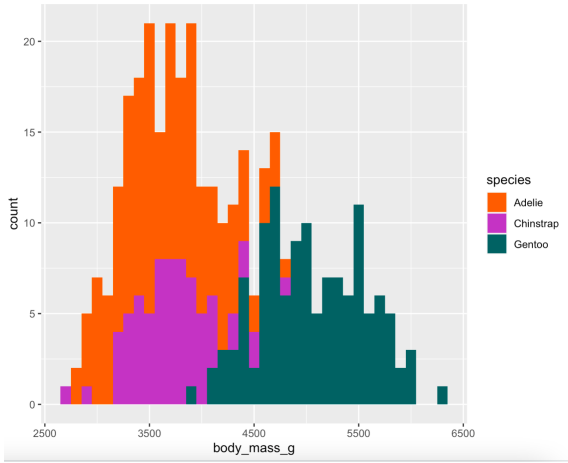
- **Trimmed sample / trimmed statistics**
  removing a certain percentage (e.g., 10%) of the values in each tail
  or removing a that are in the 99% percentile and in the 1% percentile

- **Z-scores**
  removing values that are $\pm n$ SD from the mean (z-scores, usually n = 2)

- **Box-and-whiskers plot**
  removing values that are ±1.5 IQR from the upper and lower quartiles

  - **Important**: do not remove too many outliers
    (ideally not more than 5%, and never more than 10% of the data)

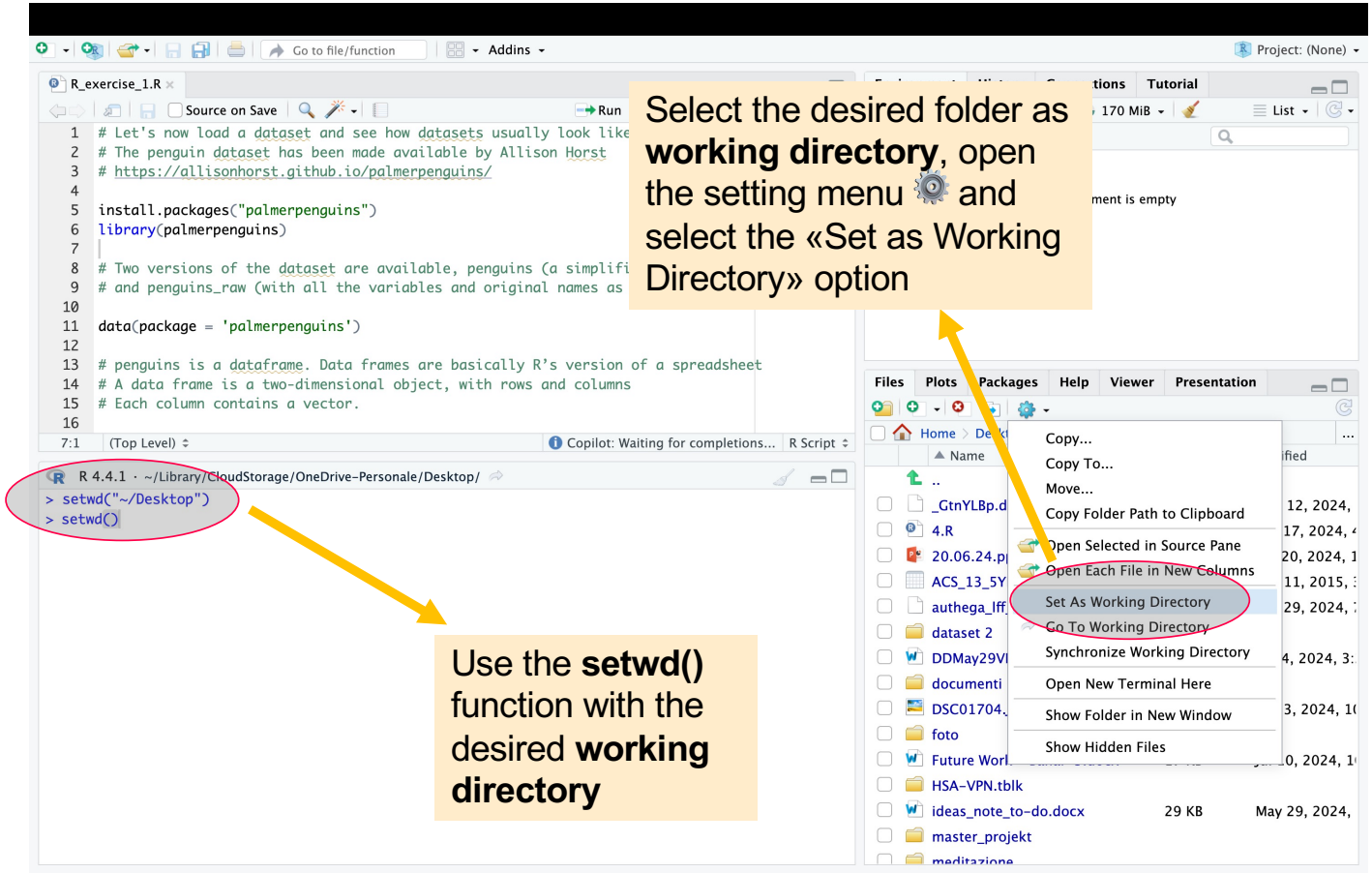
VISUALIZATIONS IN R



R_exercise_5.R & students.csv

WORKING DIRECTORY

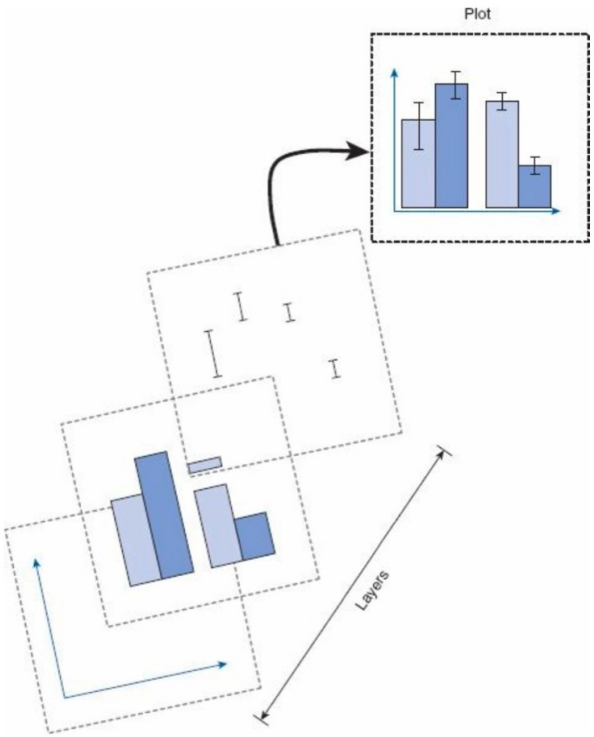
- Setting your **working directory** is important to always find the files you want to work with
- **getwd()** find the current working directory
- **setwd("..")** move back to the parent folder



VISUALIZATIONS IN R

Graphs with R: ggplot2

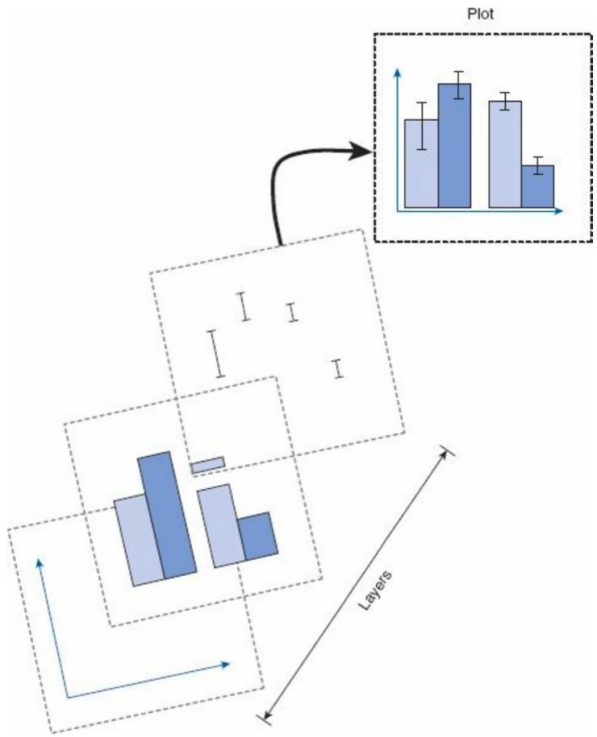
- The **ggplot2** package nowadays is a standard for creating plots and visualizations in R
- A graph in **ggplot2** is made up of a series of **layers** placed on top of each other
- You can add one **layer** after the other to add more **features** to the graph



VISUALIZATIONS IN R

Graphs with R: ggplot2

- Visual elements are known as **geoms** (short for ‘geometric objects’)
- Every time we add a **layer**, we have to tell **ggplot2** what **geom** we want displayed in it
- geoms have **aesthetic properties (aes())** that what they look like (colour, size, style and location)



VISUALIZATIONS IN R

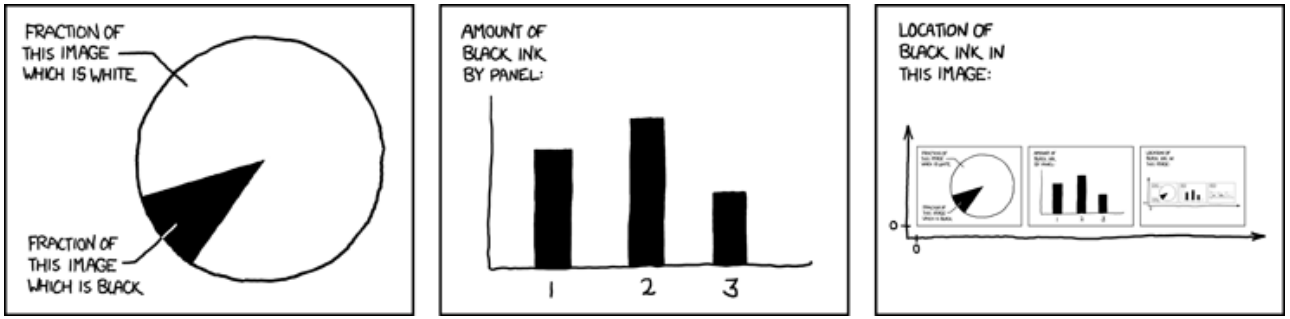
Graphs with R: ggplot2

- Open the **R_exercise_5.R** file and the **students.csv** dataset
- Load the **palmerpenguins** and the **ggplot2** packages
- Other packages you need: **languageR tydyverse, lexdec**
- Follow the comments and execute the commands in the file
- Learn how to **visualize data** with R

Questions?

Source: xkcd

HOMEWORK



- R_exercise_6.R (from Moodle)
- Watch this videos:
  - "Charts Are Like Pasta - Data Visualization Part 1" (Crash Course Statistics) https://www.youtube.com/watch?v=hEWY6kkBdpo
  - "Plots, Outliers, and Justin Timberlake: Data Visualization Part 2" (Crash Course Statistics) https://www.youtube.com/watch?v=HMkllhBI91Y